

Part 2.1: Single Motor Test

In addition to the RASCL robot, you are provided with a single motor setup. This part of the lab is for you to get familiar with ROS2 and the CiA402 Motion Controller by Faulhaber. Afterward you can adapt the methods for programming the Hardware Interface of the RASCL robot. We have provided you with a minimal Docker setup for the development, in addition to some documentation of the motion controller.

To successfully complete your task, please adhere to the following guidelines:

- Use the Docker environment for development
- Adhere to the coding guidelines published on Ilias
- Go through the documentation
- First implement the motion separately using python and the provided pysoem library
- Then implement the clock mechanism as a ros2_control hardware interface using a feed forward controller

Submission Guidelines

- Submit your files via Artemis
- Deadline: Sunday, Mai 24th, 2026, 23:59
- Deliverables:
 - Dockerfile and src files please exclude: install, build, log directories
 - Everything I need to start the container

Extra (not necessary for submission): blink the LED attached to DigOut1

Consulting hour for work package 1

There will be consulting hours for this work package on Wednesday 13th of Mai at 2pm in lab 412 (building 30.33). Please prepare dedicated questions, only these are going to be answered in person. There will not be something like a lecture.